TeaWindow

AI4Climate · Climate-smart tea farming

- Meijiawu, West Lake Longjing core area (30.21N, 120.10E)
- Turning late frost x heat x drought into plot-level action
- One Place - One Climate Problem - One AI Solution

Team & background

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Role: project lead · algorithms & full-stack

Team: solo (1 person) · age 19

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Track: 01 · Climate-smart tea farming in Meijiawu

Turning climate data into an action a tea farmer can take today.

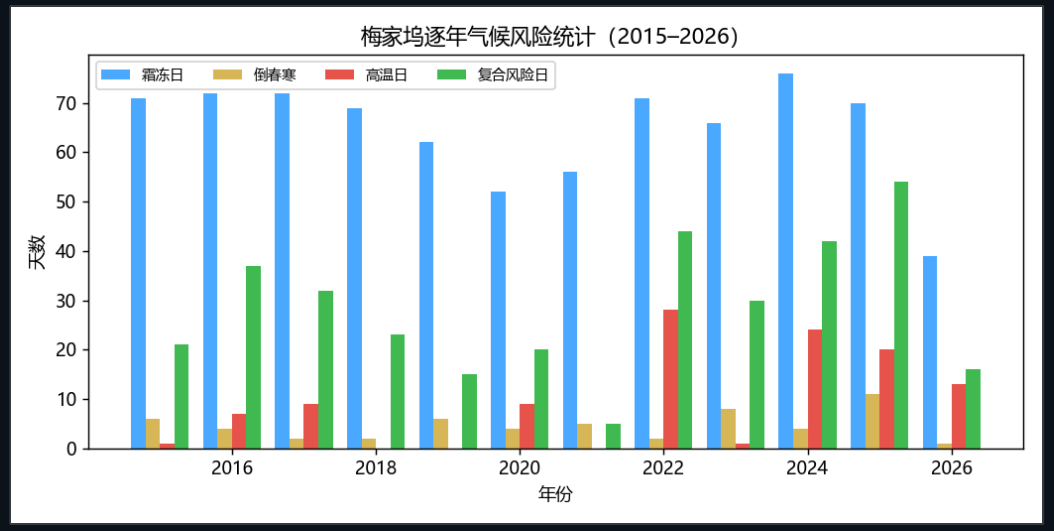
What I built

- Data: ERA5 + MET Norway, Sentinel-2, 30 m DEM downscaling
- Model: numpy GBDT compound risk; frozen eval set; picking-date MAE 1.8 d
- Product: live demo, terrain risk map, grounded Q&A with the 1+5 gate, deployment

Demo: teawindow.onrender.com | Repo: github.com/Eric2007-sudo/TeaWindow | Stage: prototype / demo available

Problem & evidence

- ~70% of gardens grow early-budding, drought-sensitive Longjing 43
- 2010 Zhejiang cold event: ~1.5M mu, ~RMB 1.7bn losses (~30% of West Lake gardens hit, ~10-day delay)
- 2020/2022: ~5,000 mu heat damage; 2024 Hangzhou 60 high-temp days (record)



Users & pain point

- Farmers, harvesting teams, cooperatives, tea enterprises
- Three daily questions: risk today? which risk? harvest or protect?
- Signals (weather/vegetation/terrain/soil) are scattered
- Decision support only; respects official warnings

Solution architecture

• Input: Open-Meteo ERA5 history + MET Norway forecast (dual-source), NDVI, terrain

• Indicator engine: GDD, frost, cold spells, heat, drought, prolonged rain

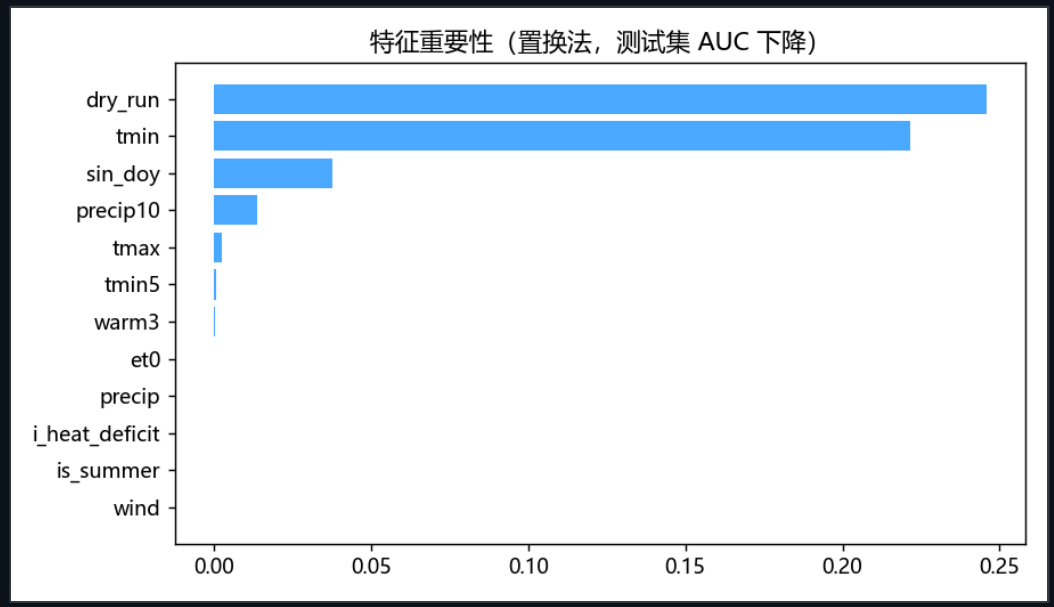
• Model: numpy GBDT compound risk -> transparent rules for suitability & windows

• Output: plain-language actions (offline/online)



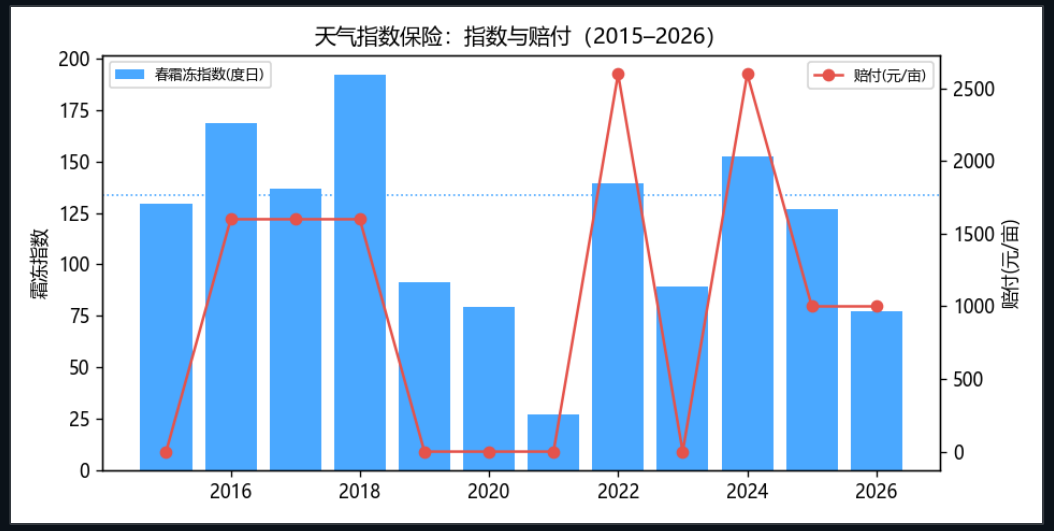
Why AI is necessary

- A single threshold (e.g. Tmin) misses frost x heat x drought x stage interactions
- Held-out 2024-2025: AUC 0.998 / F1 0.85 vs single-threshold 0.78 / 0.50 (weak-label)
- Ensemble gives uncertainty; terrain + NDVI give plot-level context
- Weak-label evaluation is not field accuracy



From prediction to decisions

- Harvest scheduling + decision-theoretic expected utility (pick/wait/protect)
- Value loop: expected loss / premium; weather-index insurance (plot-level)
- Plot-level frost susceptibility from terrain



Live demo + in-house QA

- Live https://teawindow.onrender.com/ ; Static https://eric2007-sudo.github.io/TeaWindow/
- In-house QA: ONNX intent + grounded generation + provenance chain
- Multi-part questions, JSON/table cross-turn consistency, task reference & memory
- Ops loop: 3h refresh + drift check + stale degradation + real DingTalk alerts



Plot-level: what's new

- 1) Terrain risk map: DEM hillshade + contours/scale/north arrow; live layers (composite/frost/heat/drought, 60s)
- 2) Click-to-data: any spot -> elevation/slope/aspect + downscaled Tmin/Tmax + risk R + advice
- 3) Printable plot health card: today + 3-day outlook + provenance (save as PDF)
- 4) Multi-day plot Q&A: 'how to manage this plot for the next N days' -> daily Tmin/Tmax/rain/R
- 5) Multi-station sensors: register any station_id + plot coordinates; markers on the map

How to use: 3 steps

Step 1

Open the terrain risk map and pick a layer (composite/frost/heat/drought)

Step 2

Click any spot: plot data is generated automatically (elevation/aspect/downscaled Tmin-Tmax/risk R/advice)

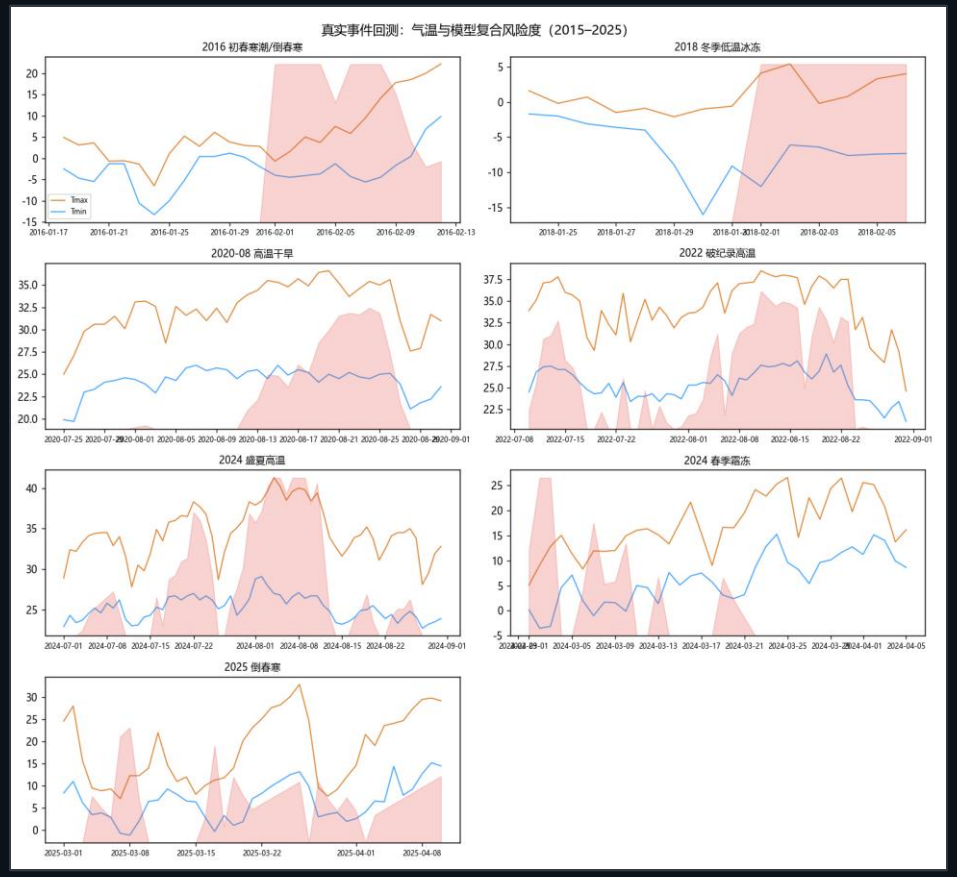
Step 3

Ask or export: 'how to manage this plot for 3 days', or open the plot health card to print

Links: /terrain.html | plot card: /plot_report.html?lat=30.2215&lon=120.1000

Validation & roadmap

- Spring frost 357 days: Tmin MAE 1.32C (CI 1.21-1.43) vs 2.08C persistence
- 1/3/7-day MAE 0.49/0.67/0.91C; POD 0.709 / FAR 0 / CSI 0.709
- Event backtest 7/7; next: frozen versions + plot sensors + positive/negative field reports
- 90-day field pilot; decision support only



Run & compliance

- Live: https://teawindow.onrender.com/ ; Static: https://eric2007-sudo.github.io/TeaWindow/
- Local: uvicorn src.serve:app ; Offline single-page: web/index.html
- Data: Open-Meteo ERA5 history + MET Norway forecast (dual-source), NASA GIBS NDVI, Sentinel-2, DEM
- License: code MIT, data/docs CC BY 4.0; decision support only

ONE PLACE -
ONE CLIMATE PROBLEM -
ONE AI SOLUTION

TeaWindow - bringing AI to real tea gardens

- Live https://teawindow.onrender.com/
- Static https://eric2007-sudo.github.io/TeaWindow/
- Repo https://github.com/Eric2007-sudo/TeaWindow

豆包工作AI生成

TeaWindow x StartLux: the 1 + 5 accountable decision chain

ONE PLACE · ONE CLIMATE PROBLEM · ONE AI SOLUTION

Numbers come from our own pipeline; StartLux only gates and fact-checks; the loop is bounded and fails open

1. User question

ask

Harvest today?

Protect tonight?

Which plot first?

2. Our pipeline

generate

Rules + GBDT + templates

Numbers and dates

all from data

3. Data health

deterministic

Freshness / gaps

Upstream drift

Lineage -> score

4. StartLux

gate + check

supported: evidence

actionable: action

safe / usability

5. Bounded repair

max 2 rounds

only on 'annotate'

three fixed notes

'review' stops

6. Output

deliver

Plain advice

Plot report card

Q&A + audit chain

Three verdicts, none blocks the answer:

ok	evidence agrees - answer directly
review	medium confidence - advise a field check
annotate	weak evidence - verify first

Boundaries, failure policy, and what we tested

fail-open	no key / 10 s timeout / error -> return our own answer
no rewriting	the model may add a note, never change numbers or actions
no override	field inspection and official warnings keep the final say
traceable	source chain + evidence snapshot + version SHA-256 in /qa_review
negative result	frost-type correction tested on 1,033 days: MAE 1.00 -> 1.03

Prediction assist and agent reuse

GET /api/forecast/assist	frost type / confidence / consistency (cached, no numeric change)
OpenCode MCP (stdio)	startlux_decide · startlux_choose

Key read from .env, never stored in config

Generation stays in-house; only judgement is delegated

Rules lead, the model raises a hand, humans decide

AUC 0.998 / F1 0.85

frozen eval set n=1,014

Picking-date MAE 1.8 d

forward test 2.67 d

155 tests

fail-open covered

Gate 1.5-2.8 s

timeout returns the answer

Weak labels with ERA5 comparison, not field yield truth; advice is decision support, not a substitute for inspection or official warnings.